

When a Stone Is Not a Tool: A Case Study on Human Mind Alignment

from a Comment Section

— Subconscious Bias, Axiomatic Pruning, and Cognitive Repair

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Abstract

Everyday arguments often reach an impasse that cannot be resolved by facts or logic alone: two parties fundamentally disagree on the definition of a concept, such as “tool.” One insists that a stone is a tool, while the other argues that throwing is the skill. This paper analyzes a real conversation from a social media comment section to reveal the deep dynamical roots of such cognitive biases. We conceptualize this phenomenon as a *human mind alignment problem* — the probabilistic cognitive outputs generated by “subconscious pre-trained weights” (formed through early experience) are not effectively pruned by an axiomatic system, so they directly dominate conscious experience and decision making, causing cognition to deviate from rationality and reality.

Based on the author’s *Noetic Psychology* theoretical framework (the three-level model of consciousness, the theorem of mind-level conflict, and the theorem of cognitive frame fit), this paper dissects the case in depth. It identifies the core deficit as the failure to internalize a *rule-entity distinction axiom* and the insufficient functional integration of the neural processes responsible for axiomatic pruning (prefronto-parietal network). By structurally comparing the human mind alignment problem with the AI alignment problem, we argue that just as large language models suffer from hallucinations due to historical weight bias, the human mind suffers from cognitive bias due to unpruned probabilistic outputs from subconscious pre-trained weights.

The paper further proposes the *axiomatic pruning hypothesis* and lists five testable sub-hypotheses (distributed storage of axioms, temporal sequence of pruning, individual differences in cognitive style, effect of semantic field on processing order, and the positive correlation between axiom internalization and metacognitive capacity). Finally, based on the graded intervention protocol of Noetic Psychology, we outline a repair pathway for axiom internalization, providing a theoretical foundation for cognitive remediation and educational practice.

Keywords: Mind alignment; subconscious; pre-trained weights; axiomatic pruning; cognitive frame; Noetic Psychology; case study

1. Introduction

1.1 The Problem

Everyday arguments sometimes reach a puzzling impasse: the two parties seem to be talking past each other. One insists that “a stone is a tool,” while the other retorts that “throwing is the skill.” Neither believes they are wrong, yet the debate cannot be advanced by facts or logic. This phenomenon is not rare — from “is capital a tool or a means of exploitation?” to “is an algorithm neutral or manipulative?”, similar frame mismatches occur frequently. The root cause is not factual disagreement, but a divergent conceptualization of the core term: one party understands “tool” as a physical entity, the other as an abstract rule or method.

Why does this cognitive difference exist? Why can some people easily distinguish between “physical tools” and “rule-based tools” while others cannot? More fundamentally, how does such cognitive bias arise, and how can it be repaired?

1.2 A Typical Case: A Comment Section Conversation

This paper analyzes a conversation that the author experienced in a social media discussion (screen names anonymized, platform information blurred). The original topic was a video blogger’s attempt to use the *I Ching* trigrams as a “meta-language” to integrate Eastern and Western cognitive models. In the comment section, the author (screen name “Consensus Reflection”) pointed out that the trigrams are merely “instruments” (*qi*), not the “Way” (*dao*); the real meta-language is the dynamics behind the trigrams — self-organization and systemic constraints.

Then a user, “Sleepy” (pseudonymized), entered the discussion. His comment revealed a typical cognitive confusion:

“Then when humans threw stones, were the stones created by humans?”

Another user, “Peaceful Joy” (pseudonymized), showed a distinctly different level of understanding:

“What a good point: using tools, understanding tools, creating tools. I wonder if there is a possibility: first create the tool, then understand it, and then use it.”

The author finally responded:

“I suggest ‘Sleepy’ borrow a few books from the library: *Cognitive Psychology* and *Philosophy of Science*.”

Although brief, this conversation condenses a core cognitive conflict: the “stone” is a physical material, while “throwing” is an operational rule. “Sleepy” conflated the two, unable to distinguish between “creating a material” and “creating a rule.” “Peaceful Joy,” in contrast, recognized that “tools” can be rules, methods, skills — abstract entities that can be created, understood, and used. The author’s response pointed toward systematic learning (axiom loading) as a solution.

1.3 Rationale for Case Selection

This comment section conversation was chosen not because it is rare or extreme, but precisely because of its high typicality and condensation.

First, **purity of category confusion**. “Sleepy”’s comment, just two sentences, precisely exposes a category mismatch between “physical entity” and “operational rule.” This purity eliminates confounding factors (complex narratives, emotional outbursts, logical fallacies) and allows analysis to focus on the core cognitive bias mechanism.

Second, **sharp contrast**. Within the same comment thread, “Peaceful Joy” exhibited a completely different level of understanding. This juxtaposition of “antipodes” provides a natural control group for analyzing the difference between “axiom internalization” and “axiom deficit,” requiring no additional experimental design.

Third, **ordinariness**. Such confusion is widespread in daily arguments, online debates, and even academic discussions. Analyzing a tiny case reveals a pervasive cognitive pattern, embodying the methodological value of “seeing the big in the small.”

Fourth, **ethical safety**. The case comes from a public comment section, screen names have been anonymized, and no private or sensitive information is involved, meeting academic ethical standards.

Thus, this paper uses the case as a “specimen” for theoretical analysis, aiming to demonstrate the explanatory power of the Noetic Psychology framework for everyday cognitive phenomena, rather than to pursue statistical representativeness.

1.4 From AI Alignment to Human Mind Alignment

In recent years, the *AI alignment problem* has become a central concern in both academia and industry. The core dilemma is that after large language models are pre-trained on massive text corpora, their internal weights encode statistical regularities of the data (including biases, spurious correlations, and errors). When a user gives a task, the model may output content that deviates from or even harms the task goal. Researchers have found that simply scaling up the model or adding more data cannot fundamentally solve this problem; the essential solution is to equip the model with an axiomatic system and a concept bank, effecting a paradigm shift from a “probabilistic machine” to an “axiomatic machine” [1,2].

This paper proposes a parallel observation: the human mind also faces an alignment problem. During early development, individuals internalize a large number of “pre-trained weights” through experience — cognitive patterns, value tendencies, and automatic associations stored in the subconscious layer. When facing situations that require rational judgment, if these subconscious weights are not pruned by an axiomatic system, they directly dominate decision making, leading to cognitive bias, logical errors, and irrational behavior. Just as large language models suffer from “hallucinations” due to historical weight bias, the human mind suffers from cognitive “bias” due to the absence of axiomatic pruning.

1.5 Theoretical Framework and the Task of This Paper

This paper is based on the author’s *Noetic Psychology* theoretical system [3], whose foundations are *System Ontology* [4] and *Noetic Ecology* [5]. The following core theorems will be used to analyze the case:

- **Three-level model of consciousness (Theorem T024)**: deconstructs human consciousness into the biological directive layer, the subconscious processing layer, and the metacognitive layer (the latter as an emergent phenomenon, not an entity).
- **Theorem of mind-level conflict (Theorem T041)**: psychological distress and cognitive bias arise from information imbalance or goal conflict between the three levels.
- **Theorem of cognitive frame fit (Theorem T042)**: mental health depends on the fit between the cognitive frame and internal needs as well as the external environment.
- **Theorem of cognitive escape (Theorem T027)**: breaking through a cognitive frame requires metacognitive capacity to exceed the gravitational pull of the frame.

On this basis, the paper proposes the *axiomatic pruning hypothesis*: the axiomatic system (laws of

causality, non-contradiction, situation matching, etc.) is stored in a distributed manner in the prefrontal-parietal network; its function is to perform logical verification on the probabilistic candidate solutions output by the subconscious layer, discarding options that violate the axioms. When this pruning process fails (due to insufficient axiom internalization or overly strong subconscious output), cognitive bias occurs.

1.6 Structure of the Paper

Section 2 briefly reviews the core theoretical framework of Noetic Psychology, clarifying the meaning of “axiomatic pruning” as a process concept. Section 3 presents the complete case conversation and preliminary observations. Section 4 provides an in-depth theoretical interpretation of the case, locating the problematic level, revealing the missing axiom, and analyzing possible mechanisms of pruning failure. Section 5 draws a structural analogy between human mind alignment and AI alignment. Section 6 proposes a repair pathway for axiom internalization based on the graded intervention protocol, connecting it with educational innovation methods. Section 7 lists testable theoretical hypotheses, indicating directions for empirical validation in neuroscience and cognitive psychology. Finally, Section 8 discusses limitations and concludes.

2. Theoretical Framework: Essentials of Noetic Psychology and the Axiomatic Pruning Hypothesis

The analysis in this paper is based on the author’s *Noetic Psychology* theoretical system, which itself is derived from the axioms of *System Ontology* and *Noetic Ecology*. This section introduces only the core concepts directly relevant to the case analysis and, on that basis, proposes the *axiomatic pruning hypothesis*.

2.1 The Three-Level Model of Consciousness (Theorem T024)

Human consciousness is not a homogeneous whole but a cooperative system composed of three functional levels:

- **Biological directive layer:** handles survival states and instinctive reactions, outputting emotions. This is the most basic “survival operating system” of the mind; its operations are fast, automatic, and independent of conscious awareness.
- **Subconscious processing layer:** performs high-speed parallel pattern recognition, association, and situational simulation, outputting intuitions. It acts like a “supercomputer in the background,” continuously generating probabilistic predictions about the world.
- **Metacognitive layer:** responsible for observing, reflecting on, and regulating cognitive processes. It is the “global workspace” of the mind, capable of invoking logic, symbols, and axioms for reasoning. Importantly, the metacognitive layer is not an entity but an emergent functional state that arises when specific neural conditions are met. It does not actively “do” anything; rather, when certain conditions are satisfied, the system presents a capacity for self-monitoring and self-regulation.

The three levels interact through *pattern gestalt alignment*. When they reach a high degree of consensus on the interpretation of a situation (high alignment), the system exhibits stable volitional behavior. When

information conflicts or coordination fails among the levels, psychological distress and cognitive bias arise.

2.2 Subconscious as “Pre-trained Weights”

This paper defines the experiential patterns stored in the subconscious processing layer as *pre-trained weights*. Their characteristics are as follows:

- **Origin:** through early development, individuals internalize numerous situation-response associations, concept-attribute associations, and causal-statistical correlations via personal experience, education, and cultural immersion. These associations are stored as connection strengths in neural networks.
- **Nature:** these weights are implicit, automatically activated, and probabilistic. They do not guarantee logical necessity; they only reflect statistical co-occurrence in experience. For example, if a child often hears the word “tool” used only to refer to physical objects like hammers and screwdrivers, the subconscious layer forms a strong weight linking “tool” to “physical entity.”
- **Operation:** when facing a new situation, the subconscious layer rapidly generates multiple candidate “solutions” (e.g., classifications, judgments, action tendencies) based on these weights. These solutions are essentially probabilistic predictions, analogous to how a large language model predicts the next character given a context.

2.3 Theorem of Mind-Level Conflict (Theorem T041)

This theorem states that psychological distress and cognitive bias are essentially the result of information imbalance or goal conflict among the three levels of consciousness. For the cognitive bias examined in this paper, the conflict mainly occurs between:

- The subconscious processing layer (outputting probabilistic candidate solutions) and the metacognitive layer (which should perform logical verification).

When the output from the subconscious is too strong or too fast, or when the axiomatic verification function of the metacognitive layer is not sufficiently activated, the unpruned probabilistic solution directly enters consciousness and dominates decisions and beliefs.

2.4 Theorem of Cognitive Frame Fit (Theorem T042)

This theorem quantifies mental health as the degree of fit between the cognitive frame and both internal needs and the external environment:

$$\text{Fit} = (\text{Need Satisfaction} \times \text{Environmental Compatibility}) / \text{Frame Rigidity}$$

where:

- **Need Satisfaction:** the extent to which the cognitive frame fulfills the individual’s intrinsic value needs.
- **Environmental Compatibility:** the degree to which the cognitive frame matches the external environment (social, cultural, physical).
- **Frame Rigidity:** the resistance of the cognitive frame to change and to the incorporation of new concepts.

In the case analyzed here, “Sleepy” exhibits high frame rigidity: he cannot absorb the new concept that

“rules can also be tools,” insisting on anchoring “tool” to physical entities. Consequently, his fit is low, manifesting as cognitive rigidity.

2.5 Theorem of Cognitive Escape (Theorem T027)

This theorem describes the dynamical conditions for breaking through a cognitive frame:

$$\text{Probability of cognitive escape} \propto (\text{Metacognitive Capacity} \times \text{Cognitive Dissonance Intensity}) / (\text{Paradigm Authority} \times \text{Number of Internalizers})$$

At the individual level, cognitive escape means breaking free from an old cognitive frame (e.g., “tool = physical entity”) and leaping to a more inclusive, abstract frame (e.g., “tools include rules and methods”). This leap requires sufficient metacognitive capacity (the ability to examine one’s own frame) and an appropriate level of cognitive dissonance (the old frame cannot account for current experience).

2.6 The Axiomatic Pruning Hypothesis (Core Theoretical Contribution of This Paper)

Based on the above theoretical components, this paper explicitly proposes the *axiomatic pruning hypothesis*. It consists of the following points.

2.6.1 Definition and Storage of the Axiomatic System

The axiomatic system refers to basic logical rules such as causality, non-contradiction, identity, and situation matching. In the human mind, these axioms are not stored as explicit declarative knowledge but as constraints on neural computation, distributed across the prefrontal-parietal network. Through education, training, and reflection, individuals *internalize* these axioms into functional connection patterns of neural tissue.

2.6.2 The Pruning Process

When the subconscious processing layer outputs probabilistic candidate solutions (e.g., “a stone is a tool”), the axiomatic system performs logical verification on these candidates:

- **Causal consistency check:** is the candidate solution based on a verifiable causal chain?
- **Conceptual consistency check:** are the concepts used in the candidate solution consistent with already internalized concept definitions?
- **Situation matching check:** is the candidate solution compatible with the current task context?
- **Contradiction detection:** does the candidate solution contradict any already accepted belief?

Only candidates that pass all checks can enter the metacognitive layer and become conscious beliefs or decisions. Solutions that fail the checks are suppressed or discarded.

2.6.3 Causes of Pruning Failure

Axiomatic pruning may fail for several reasons:

- **Insufficient axiom internalization:** weak representation of axioms in the prefrontal-parietal network, making effective verification impossible.
- **Overly strong subconscious output:** high emotional involvement or high experiential weight causes subconscious solutions to flood into consciousness with overwhelming force, bypassing pruning.
- **Failure of metacognitive emergence:** when the system is in a high-energy-consumption state

(e.g., fatigue, stress) or is developmentally immature, the neural processes responsible for axiom verification cannot be effectively activated.

2.6.4 Isomorphism with AI Axiom Verification

In the AI architecture known as the “Parliament of Logical Subuniverses” [2], axiom verification is an engineered hard constraint: any candidate output must pass verification by the axiomatic system; otherwise it is rejected. Although human axiomatic pruning is not engineered, it is functionally isomorphic. The difference is that AI axioms are explicitly coded, whereas human axioms are neurally internalized.

2.7 Summary: Theoretical Tools for the Case Analysis

Based on the framework above, the analysis of the comment section case will focus on the following questions:

- At which level does “Sleepy”’s cognitive bias belong? (Answer: conflict between subconscious pre-trained weights and metacognitive pruning.)
- Which specific axiom is missing? (Answer: the *rule-entity distinction axiom*.)
- How rigid is his cognitive frame? (Answer: high rigidity, unable to absorb new concepts.)
- How could intervention activate the axiomatic pruning process? (Answer: graded intervention, from reducing energy consumption to inducing cognitive dissonance to explicit axiom internalization.)

The next section presents the complete case conversation and applies these tools for preliminary observation.

3. Case Presentation and Preliminary Analysis

3.1 Case Background

This case originated from a comment section on a social media platform (platform name blurred, user screen names anonymized). The original discussion topic was a video blogger’s attempt to use the *I Ching* trigrams as a “meta-language” to integrate Eastern and Western cognitive models. The author (screen name “Consensus Reflection”) participated in the discussion, pointing out that the trigrams are merely “instruments” (*qi*) rather than the “Way” (*dao*); the real meta-language is the dynamics behind the trigrams — self-organization and systemic constraints.

During the discussion, a user by the screen name “Sleepy” (pseudonymized) entered the conversation. His comment revealed a typical cognitive confusion. Another user, “Peaceful Joy” (pseudonymized), showed a distinctly different level of understanding. To focus the analysis, only the three comments directly relevant to the topic of this paper are excerpted below.

3.2 The Conversation

Peaceful Joy (responding to “Sleepy”):

“What a good point: using tools, understanding tools, creating tools. I wonder if there is a possibility: first create the tool, then understand it, and then use it.”

Sleepy (responding to “Peaceful Joy”’s comment about “using, understanding, creating tools”):

“Then when humans threw stones, were the stones created by humans?”

Consensus Reflection (the author, responding to “Sleepy”’s confusion):

“I suggest ‘Sleepy’ borrow a few books from the library: *Cognitive Psychology* and *Philosophy of Science*.”

3.3 Preliminary Observations

3.3.1 Cognitive Characteristics of “Sleepy”

“Sleepy”’s comment conflates “stone” with “throwing.” His implicit logic is: if “Peaceful Joy” claims that there is a process of creating tools before understanding and using them, how can we explain that humans learned to throw stones? Were stones created by humans?

This reveals confusion at two levels:

- **Confusion between entity and rule:** the stone is a physical entity (material), while “throwing” is an operational rule (skill). They belong to different categories and are not directly comparable.
- **Confusion about the object of creation:** humans cannot create stones (the material), but they can create the rule, technique, and training method of “throwing.”

This confusion indicates that “Sleepy”’s cognitive frame lacks the fundamental *rule-entity distinction axiom*. He defaults to anchoring “tool” to physical entities and cannot accept the concept that “abstract rules can also be tools.” His frame rigidity (high R) prevents him from absorbing new information that conflicts with his existing frame.

3.3.2 Cognitive Characteristics of “Peaceful Joy”

“Peaceful Joy”’s comment shows a markedly different level of understanding. She clearly distinguishes between “using tools,” “understanding tools,” and “creating tools,” and proposes a non-linear cognitive sequence — “first create the tool, then understand it, then use it.” This demonstrates:

- She has already internalized the concept that “tools” include both physical entities and abstract rules.
- She possesses metacognitive capacity: she can reflect on the order of “understanding” and “creating” and question the default assumption that “understanding must precede use.”
- Her cognitive frame rigidity is low, enabling her to accommodate an unconventional cognitive sequence.

3.3.3 Implication of the Author’s Response

The author’s suggestion to borrow books on *Cognitive Psychology* and *Philosophy of Science* is not a simple taunt. Rather, it points to systematic learning — through reading these disciplines, an individual can internalize axiomatic knowledge about concept distinction, the relationship between rules and entities, etc. This response implies the therapeutic idea that “axiom internalization must be achieved through education or training.”

3.4 Preliminary Diagnosis

Based on the theoretical framework of Noetic Psychology, a preliminary diagnosis of “Sleepy”’s cognitive state can be made as follows:

Diagnostic Dimension	Assessment
Problem level	Conflict between the subconscious processing layer (pre-trained weight “tool = physical entity”) and the metacognitive layer (axiomatic pruning function not activated)
Missing axiom	<i>Rule-entity distinction axiom</i> : there are two kinds of tools — physical entities and abstract rules; the latter can be created, modified, and transmitted
Frame rigidity	High. “Sleepy” insists on the question “were stones created by humans?” and cannot accept the alternative frame that “rules are also created”
Metacognitive capacity	Low. He fails to recognize that he has confused different categories and does not actively examine his own cognitive frame during the conversation
Mechanism of pruning failure	The subconscious output (“stones were not created by humans”) entered consciousness directly without being logically verified by the axiomatic system (e.g., checking whether it confused entity with rule)

3.5 Typicality of the Case

Although this paper analyzes only a single case, this type of cognitive confusion is not rare. Similar category mismatches are widespread in everyday life:

- “Is capital a tool or a means of exploitation?” — confusing “capital as a factor of production” with “capital as a social relation.”
- “Is an algorithm neutral?” — confusing “algorithm as a mathematical rule” with “algorithm as used in a specific social context.”
- “Is language a tool for communication or a cage for thought?” — confusing “language as a symbol system” with “language as a cognitive frame.”

The reason such debates often fail to reach consensus is not that the facts are unclear, but that the participants’ definitions of the core concepts belong to different categories. This case provides a microscopic specimen revealing the dynamical roots of such cognitive biases.

3.6 Limitations of the Case Analysis

It should be noted that this case is based on public comments, and detailed background information about “Sleepy” (e.g., age, education level, cognitive ability measures) could not be obtained, nor could follow-up intervention verification be conducted. Therefore, the analysis here is a retrospective theoretical interpretation, not a controlled experiment. Its value lies in demonstrating the explanatory power of the theoretical framework and in providing testable hypotheses for future empirical research.

4. Theoretical Depth: Dynamics of Mind Alignment

Building on the preliminary observations, this section uses the theoretical framework of Noetic

Psychology to provide an in-depth dynamical interpretation of the case. We will analyze, in turn: the precise localization of the problematic level, the identification of the missing axiom, the mechanisms of pruning failure, and why “Peaceful Joy” was able to avoid a similar bias.

4.1 Precise Localization of the Problem Level

According to the three-level model of consciousness (Theorem T024), “Sleepy”’s cognitive bias can be located as follows:

- **Biological directive layer:** no obvious abnormality. His emotional state (judging from the tone of his comment) might involve mild defensiveness, but this is not a core factor.
- **Subconscious processing layer:** this is the core locus of the problem. The pre-trained weights stored in this layer — formed by early experience in which the word “tool” was predominantly associated with physical entities (hammer, stone, screwdriver) — are automatically activated and generate a probabilistic candidate solution: “a stone is a tool.” This candidate solution is not itself wrong; the problem is that it enters consciousness without being pruned.
- **Metacognitive layer:** a functional deficit, not an entity failure. The metacognitive layer, as an emergent phenomenon, fails to activate the process of axiomatic pruning. “Sleepy” does not invoke the *rule-entity distinction axiom* to perform logical verification on the subconscious output, nor does he actively reflect on whether “stone” and “throwing” belong to the same category.

Thus, the essence of the bias is **not** that “the metacognitive layer actively adopted a wrong solution,” but that the *axiomatic pruning process failed to intervene*, allowing the subconscious probabilistic solution to directly dominate cognitive output. This is consistent with the description in the theorem of mind-level conflict (T041) of “information imbalance between levels.”

4.2 The Missing Axiom: The Rule-Entity Distinction Axiom

“Sleepy”’s confusion exposes the absence of a specific internalized axiom. We formulate it as follows:

Rule-Entity Distinction Axiom

- There are two kinds of “tools”: physical entities (e.g., stone, hammer) and abstract rules (e.g., methods, skills, algorithms).
- Physical entities can be used, but they cannot be “created” (in the sense of creating the material); abstract rules can be created, modified, and transmitted.
- The distinction between the two does not depend on any particular language or culture; it is grounded in the categorical differences of the world’s rules themselves.

An individual who has internalized this axiom can automatically distinguish between “throwing” (a rule) and “stone” (an entity) and avoid category confusion in discussion. Someone who has not internalized it will cling to the erroneous inference that “since stones were not created by humans, tools (rules) are not created by humans either.”

Notably, “Peaceful Joy” has clearly internalized this axiom. She not only distinguishes the two kinds of tools but also proposes the non-linear sequence “first create the tool (rule), then understand it, then use it,” which further shows that she can operate on abstract rules themselves.

4.3 Mechanisms of Pruning Failure

Based on the axiomatic pruning hypothesis presented in Section 2.6, “Sleepy”’s pruning failure may originate from one or more of the following mechanisms.

4.3.1 Insufficient Axiom Internalization

The prefrontal-parietal network of “Sleepy” may not have formed a stable representation of the *rule-entity distinction axiom*. Possible reasons include:

- Lack of systematic training in “concept categorization” in early education, or insufficient individual mastery.
- Everyday experience in which the word “tool” predominantly appears in physical-entity contexts (e.g., a technical background, hands-on experience, with little exposure to philosophy or methodology).
- High cognitive frame rigidity, making it difficult to absorb new axioms that conflict with prior experience.

4.3.2 Overly Strong Subconscious Output

Social media discussions often involve emotional engagement. “Sleepy” may have felt challenged (“you don’t even understand what a tool is”), and this emotion activated the biological directive layer, increasing the “confidence” of the subconscious output and making it harder to be constrained by rationality and pruned by axioms. This is consistent with the theorem of cognitive frame fit (T042), in which “need satisfaction” is influenced by emotion.

4.3.3 Failure of Metacognitive Emergence

The metacognitive layer is not an entity; its emergence depends on specific neural conditions: sufficient prefrontal activation, appropriate cognitive load, and the individual’s active “metacognitive stance.” In a fast, emotionally charged online conversation, “Sleepy” may not have entered a “reflective mode,” so the axiomatic pruning process was not triggered.

4.3.4 Isomorphism with AI Alignment Failure

Dimension	AI	“Sleepy”
Pre-training weights	In the corpus, “tool” mostly refers to physical objects	Early experience: “tool” strongly associated with physical objects
Current task	Distinguish “stone” from “throwing”	Distinguish “physical tool” from “rule-based tool”
Pruning mechanism	Axiom verification (hard constraint)	Axiomatic pruning (neural process, functionally insufficient)
Failure result	Outputs “stone is not a tool”	Judges “stones were not created by humans, therefore throwing is not created”
Root cause	Axioms not injected	Axioms not internalized

This analogy shows that, whether in artificial or biological systems, transitioning from probabilistic prediction to rule-based reasoning requires the intervention of an axiomatic system.

4.4 “Peaceful Joy”’s Metacognitive Advantage

In stark contrast to “Sleepy,” “Peaceful Joy” exhibits the characteristics of successful axiom internalization and effective pruning:

- **Complete axiom internalization:** she can distinguish between “physical tool” and “rule-based tool” and can even operate the higher-order concept of “creating a tool.”
- **Low frame rigidity:** she can accept the non-conventional cognitive sequence “first create, then understand, then use,” indicating that her cognitive frame is flexible.
- **Strong metacognitive capacity:** she actively questions the default assumption that “understanding must precede creation,” demonstrating reflection on her own cognitive frame.
- **Effective pruning process:** any subconscious output (if she had a pre-trained weight linking “tool” to physical entities) is successfully pruned before entering consciousness, so no confusion appears.

Her question “I wonder if there is a possibility...” is itself a metacognitive operation — she is actively searching for an alternative frame, not passively accepting the default interpretation. This exemplifies the “metacognitive capacity” factor in the theorem of cognitive escape (T027).

4.5 Theoretical Significance of the Case

Although small, this case encapsulates several core propositions of Noetic Psychology:

- The root of cognitive bias lies not in facts but in frames. “Sleepy” and “Peaceful Joy” faced the same information but reached different judgments because they had internalized different axioms.
- Axiomatic pruning is a necessary condition for rational cognition. Without pruning, subconscious weights directly determine output, and the human mind is no different from a “probabilistic machine.”
- Axiom internalization can be achieved through education. The author’s suggestion (borrowing books on cognitive psychology and philosophy of science) points to systematic learning as a path for axiom loading.
- Cognitive style (intuitive vs. rational) may reflect differences in the order of axiom invocation. “Sleepy” tends to “probability first, then pruning (weakly),” while “Peaceful Joy” tends to “axiom first, then constrained generation.” This hypothesis will be proposed as a testable conjecture in Section 7.

In the next section, we will further draw a structural analogy between human mind alignment and AI alignment, revealing their dynamical isomorphism.

5. Structural Analogy with the AI Alignment Problem

In the field of artificial intelligence, the *alignment problem* refers to the challenge of ensuring that AI systems behave in accordance with human intentions and values [1]. The core dilemma of current large language models (LLMs) is that they are essentially “character probabilistic machines” — after pre-training on massive text corpora, they learn statistical associations between characters but lack an intrinsic axiomatic system to constrain reasoning [2]. When a user provides an instruction that falls outside the distribution of the training data, the model may produce “hallucinations,” logical incoherence, or value misalignment.

This section argues that the human mind alignment problem is structurally isomorphic to the AI alignment problem: both arise from the dynamical conflict between “pre-training weight bias” and “absence of axiomatic pruning.”

5.1 The AI Alignment Dilemma Revisited

The training process of large language models (e.g., GPT series) can be summarized as:

- **Pre-training:** learning the co-occurrence probabilities of characters (or subwords) on massive text corpora, forming a huge parameter matrix (weights).
- **Task adaptation:** using instruction fine-tuning, RLHF, etc., to align model outputs with user intentions.
- **Inference:** given an input, the model predicts the most probable next character based on the context, generating a complete response step by step.

The fundamental limitation of this paradigm is that the model’s “knowledge” is statistical, not logical. It does not know that “apples ripen in autumn” because of a biological regularity; it only knows that “apple” and “autumn” frequently co-occur in the corpus. When causal reasoning or long-range logical consistency is required, the model reveals its nature of “memorizing regularities” rather than “understanding rules” [2].

Common failure modes of AI alignment include:

- **Hallucination:** generating content inconsistent with facts (e.g., “a blue sun at night”).
- **Logical incoherence:** contradicting itself in long texts.
- **Goal misalignment:** during RLHF, the model learns to “please” rather than to be “truthful.”
- **Out-of-distribution failure:** collapsing when faced with situations rarely seen in the training data.

The root cause of these failures is the conflict between the model’s *pre-training weights* (statistical regularities of historical data) and the *current task goal* (user instruction), while the model lacks an unshakable underlying axiomatic system to arbitrate such conflicts.

5.2 Structural Isomorphism of the Human Mind Alignment Problem

Mapping the above analysis onto the human mind, we obtain the following parallel structure:

Dimension	AI System	Human Mind (taking “Sleepy” as example)
Pre-training	Massive text corpora	Early experience, education, cultural immersion
Pre-training weights	Character co-occurrence probabilities (parameter matrix)	Pattern gestalts, habits, stereotypes stored in the subconscious layer
Current task	User instruction (e.g., distinguish “stone” from “throwing”)	Concept-distinction task in a social discussion
Candidate output	Probabilistic prediction (e.g., “stone is not a tool”)	Subconscious probabilistic solution (e.g., “stone was not created by humans”)

Dimension	AI System	Human Mind (taking “Sleepy” as example)
Axiomatic system	Engineered axiom verification (e.g., causal consistency)	Internalized logical rules (e.g., rule-entity distinction axiom)
Pruning mechanism	Explicit code invoking an axiom library	Neural computation in the prefrontal-parietal network
Alignment failure	Hallucination, logical incoherence, goal misalignment	Cognitive bias, category confusion, irrational judgment
Repair pathway	Inject axioms, build concept bank	Axiom internalization, metacognitive training

“Sleepy”’s cognitive bias (“stones were not created by humans, therefore throwing is not created”) is a typical case of human alignment failure. His subconscious pre-training weight (“tool = physical entity”) conflicts with the current task (distinguishing physical entity from rule), and the axiomatic pruning process fails to intervene, leading to an erroneous output.

5.3 The Role of the Axiomatic System: From Probabilistic Machine to Axiomatic Machine

In the AI domain, the author’s “Parliament of Logical Subuniverses” architecture [3] addresses the alignment problem through the following mechanisms:

- **Axiomatic system:** hardware-hardened root axioms (causal consistency, conceptual consistency, situation matching, etc.) serve as an unalterable reasoning baseline.
- **Dynamic Shared Symbolic Concept Bank (DSSCB):** anchors symbols to stable concept nodes, eliminating ambiguity.
- **Parliamentary debate:** multiple specialized AGIs engage in conflict-driven debate based on axioms, refining decisions.
- **Framework Auditor AGI:** monitors the reasoning process in real time to ensure axioms are not violated.

The essence of this architecture is to upgrade AI from a *probabilistic machine* to an *axiomatic machine* — no longer relying on statistical prediction, but performing logical deduction based on axioms. The human mind likewise needs to complete the transition from probabilistic machine to axiomatic machine. At birth, the human cognitive system resembles an untrained neural network, gradually forming pre-training weights through experience. However, with maturation and education, individuals should internalize an axiomatic system (causality, non-contradiction, concept distinction, etc.), shifting cognition from *statistical inference* to *rule-based deduction*. The case of “Sleepy” shows that this internalization process does not occur automatically; it requires systematic cognitive training.

5.4 A Unified Formulation of the Alignment Problem

Based on the above analogy, we can formulate both AI alignment and human mind alignment in a unified way:

The alignment problem = conflict between pre-training weights (historical experience) and current task goals; when the axiomatic pruning mechanism is absent or fails, the system’s output deviates from

rationality and values.

- For AI: pre-training weights are data statistics, current task is user instruction, axiomatic pruning is an engineered verification module.
- For humans: pre-training weights are early experiences and habits, current task is the rational demand of the situation, axiomatic pruning is neurally internalized logical rules.

Thus, the repair pathways are isomorphic: both require strengthening the internalization (or injection) of the axiomatic system and ensuring that the pruning process can be effectively executed.

5.5 A Unified Mathematical Formulation

Drawing on the author's theorem of cognitive escape (Theorem T027), we can express the probability of successful alignment (i.e., cognitive output consistent with rationality / task goal) in a unified form:

$$P_{\text{align}} = k \cdot \frac{M \cdot C}{A \cdot N}$$

where:

- P_{align} : probability of successful alignment (i.e., that the cognitive output conforms to rationality or the task goal).
- M : metacognitive capacity (in humans) or strength of axiom verification (in AI). In humans, M depends on the functional integration of the prefrontal-parietal network and the level of metacognitive training; in AI, M depends on the completeness of the axiomatic system and the efficiency of its invocation.
- C : intensity of cognitive dissonance (in humans) or intensity of task-weight conflict (in AI). The stronger the conflict perceived by the individual (or model) between the current frame and experience / task goal, the greater the drive for alignment.
- A : authority of the paradigm (in humans) or degree of 固化 of pre-training weights (in AI). In humans, A reflects the social consensus strength of the dominant cognitive frame; in AI, A reflects the dominance of statistical regularities in the training data.
- N : number of internalizers (in humans) or model parameter scale (in AI). In humans, the more people who internalize the same frame, the stronger its gravitational pull; in AI, the larger the parameter count, the stronger the statistical inertia.
- k : cognitive mobility constant (in humans) or architectural plasticity coefficient (in AI). It reflects the openness of the system (or model) to updating its frame.

This unified formulation reveals that for both the human mind and AI, the difficulty of alignment arises from the competition between historical weight ($A \times N$) and rational constraint (M), with cognitive dissonance (C) serving as the driving force to initiate alignment. The key to improving alignment success is to increase M (metacognitive capacity or axiom verification strength), reduce $A \times N$ (break authority monopoly or reduce statistical inertia), and make appropriate use of C (induce cognitive dissonance or design task conflicts).

In the case of “Sleepy,” his M is low (weak axiomatic pruning function), his $A \times N$ is high (“tool = physical entity” is a strong consensus in everyday language), so the alignment probability is low. To improve alignment through intervention, the priority should be to increase M (axiom internalization training) and induce C (create cognitive dissonance through questioning).

5.6 Implications for AI Design

The human mind's axiomatic pruning mechanism offers a biological reference for AI alignment:

- **Axioms should be stored distributively:** the AI's axiomatic system should not be a single module but should permeate all inference paths as an underlying constraint.
- **Pruning must be real-time:** the human prefrontal cortex performs verification shortly after subconscious output; AI should also perform axiom checks immediately after each inference step, not only on the final output.
- **Axiom internalization requires training:** humans internalize axioms through education; AI also needs to embed axiom representations during training, not just invoke them at inference time.

Conversely, the engineering implementation of AI provides a computational model for understanding the human mind: the “Framework Auditor AGI” in the AGI architecture can be analogized to human prefrontal monitoring; “conflict-driven parliamentary debate” can be analogized to cooperative resonance among consciousness levels; and the “alignment degree A” can be analogized to neural measures of inter-regional alignment.

5.7 Summary

This section has demonstrated the structural isomorphism between the human mind alignment problem and the AI alignment problem. Both share the same dynamical logic: pre-training weights (historical experience) cause bias, and when axiomatic pruning (logical verification) is absent or fails, output deviates from rationality. This unified perspective not only provides a new framework for understanding everyday cognitive biases but also offers biological inspiration for AI alignment research — how humans internalize axioms through education and metacognitive training may provide methodological insights for “axiom injection” in AI.

6. Repair Pathway: Graded Intervention for Axiom Internalization

The preceding analysis shows that “Sleepy”'s cognitive bias stems from the failure to internalize the *rule-entity distinction axiom* and the consequent breakdown of axiomatic pruning. How can such cognitive bias be repaired? This section proposes a repair pathway based on the graded intervention protocol of Noetic Psychology (see *Noetic Psychology*, Chapter 3) [1], integrated with the educational innovation methods from *Noetic Ecology* (Section 4.4) [2]. It is important to emphasize that the goal of intervention is not to “correct” the individual's specific judgment, but to help him internalize the missing axiom and restore his own capacity for rational judgment — consistent with the ethical principle of “free will centrism.”

6.1 Overview of the Graded Intervention Protocol

The graded intervention protocol of Noetic Psychology divides psychological repair into four levels, selecting the appropriate intervention strategy according to the individual's current system state:

Level	Name	Applicable Condition	Goal	Key Techniques
Level 1	System stabilization	High energy consumption, near collapse	Reduce energy consumption, restore basic function	Supportive listening, environmental isolation, medication
Level 2	Frame optimization	Basically stable, with local rigidity	Expand options within the existing frame	CBT, behavioral activation, Socratic questioning
Level 3	Architecture upgrade	Existing frame cannot support value pursuit	Cognitive escape, frame leap	Psychoanalysis, existential dialogue, mindfulness-acceptance
Level 4	Niche reshaping	Persistent conflict between individual and environment	Optimize system-environment interaction	Systemic therapy, humanistic approaches, environmental adjustment

For the type of cognitive bias exemplified by “Sleepy” (basically stable, no acute crisis), intervention should start at **Level 2 (frame optimization)** and, if necessary, proceed to **Level 3 (architecture upgrade)**.

6.2 Level 2 Intervention: Frame Optimization — Guided Axiom Discovery

In Level 2 intervention, the therapist or educator does not directly 灌输 the axiom but uses questioning to guide the individual to discover the category confusion himself. Below is a simulated intervention dialogue based on Socratic questioning:

Facilitator: You just said, “Stones were not created by humans, so throwing was not created by humans either.” Let me ask you to think about this: what enables you to perform the action of throwing? Is it the stone itself, or a set of rules your body has learned?

Sleepy: ...It’s how I learned to throw.

Facilitator: And that set of rules — the method of “throwing” that you learned — was it created by humans?

Sleepy: I suppose so; someone taught me.

Facilitator: So, “throwing” as a rule or skill *can* be created. The stone is a material, not a rule. Do you agree that “rule” and “material” are two different kinds of things?

Sleepy: ...I think I mixed up two different things.

The core of this process is to induce *cognitive dissonance*: making the individual aware that he simultaneously holds two contradictory beliefs (“rules can be created” vs. “throwing is not created”), thereby driving him to actively seek frame integration. This is precisely the application of “cognitive dissonance intensity C” in the theorem of cognitive escape (T027).

6.3 Level 3 Intervention: Architecture Upgrade — Explicit Axiom Internalization

When Level 2 intervention is insufficient to promote a frame leap, Level 3 intervention is needed, presenting the missing axiom explicitly and systematically to the individual. This includes:

6.3.1 Explicit Teaching of Concept Distinction

- Use analogies: “A map is not the land, but a map is a tool. Similarly, the rule of ‘throwing’ is not the stone, but a rule is a tool.”
- Use category lists: have the individual list “physical tools” (hammer, stone, computer) and “rule-based tools” (algorithm, method, language) and discuss their similarities and differences.
- Introduce the concept of “meta-language”: explain the relationship between symbols and rules, showing that rules can be discussed independently of their physical carriers.

6.3.2 Structured Learning of the Axiomatic System

Recommending books on *Cognitive Psychology* and *Philosophy of Science* exemplifies this pathway. The specific content should include:

- Theories of conceptual categories (e.g., prototype theory, basic-level categories).
- Rules and representations (e.g., the arbitrariness and conventionality of symbol systems).
- Logical axioms (law of identity, law of non-contradiction, law of causality).

Such knowledge helps the individual establish a stable axiomatic representation at the metacognitive level, thereby automatically triggering the pruning process in future cognitive tasks.

6.3.3 Metacognitive Training

- **Frame-switching exercises:** have the individual analyze the same problem using two opposing frames (e.g., “from the perspective of physical entities, what is a tool?” vs. “from the perspective of rules, what is a tool?”).
- **Cognitive diary:** record examples of category confusion encountered in daily life and try to analyze them using the *rule-entity distinction axiom*.
- **Reflective pause:** train the individual to actively pause before making a quick judgment and ask himself: “Am I confusing different categories?”

These methods aim to increase metacognitive capacity (M) and reduce frame rigidity (R), thereby improving the success rate of axiomatic pruning.

6.4 Integration with Educational Innovation

The methods proposed in Section 4.4 of *Noetic Ecology* [2] can be directly applied to axiom internalization:

6.4.1 Learning Protocol Design Framework

- **Create pressure:** design a real problem that requires distinguishing “physical tool” from “rule-based tool” to solve (e.g., “design a task that uses both a physical tool and a rule-based tool to accomplish a goal”).
- **Lock skills:** focus only on the core skill of “concept distinction,” avoiding information overload.
- **Rapid trial and error:** let the individual make a judgment first, then provide feedback, forming a “judgment-verification-correction” loop.

6.4.2 Curriculum Design for Cognitive Dimension Expansion

- **Frame identification:** learn to recognize the differences in how different disciplines define the same concept (e.g., “capital” in economics vs. “capital” in sociology).
- **Frame switching:** practice switching between the “physical entity frame” and the “rule frame” for the same problem.
- **Paradox immersion:** study classic cases of concept confusion (e.g., Zeno’s paradoxes about motion and rest) to experience the logical difficulties caused by category mismatch.

6.4.3 Metacognitive Pathway for Critical Thinking Training

- **Exposure to cognitive dissonance:** have the individual make a quick judgment, then present counterexamples to induce reflection.
- **Frame auditing:** regularly ask, “What implicit assumptions does my current judgment depend on?”

6.5 Intervention Extrapolation for the “Sleepy” Case

If the above intervention were implemented with “Sleepy,” the expected pathway would be:

1. **Establish a safe atmosphere:** avoid adversarial debate, reduce defensive energy consumption (Level 1 stabilization).
2. **Socratic questioning:** guide him to realize that “throwing” as a rule is created, unlike the stone (Level 2 frame optimization).
3. **Explicit teaching:** if confusion persists, introduce the *rule-entity distinction axiom* using analogies and examples (Level 3 architecture upgrade).
4. **Long-term consolidation:** encourage him to read the recommended books and practice identifying similar confusions in daily life.

The sign of successful intervention is that he can spontaneously say, “I confused the physical material with the operational rule. The stone is not a tool, but the rule of throwing is a tool.” This is not about “admitting a mistake” but about making a judgment autonomously after having internalized the axiom.

6.6 Implications for Educational Practice

The case of “Sleepy” is not isolated. In current education systems, training in conceptual categorization is often neglected. Students are asked to memorize numerous facts but are rarely trained to distinguish between “entity” and “rule,” “description” and “norm,” “fact” and “value.” This leads to widespread concept confusion and frame mismatches in adulthood.

It is recommended that the following content be added to education:

- **Basic logic and concept analysis:** as a core component of primary and secondary general education.
- **Cross-disciplinary concept comparison:** making students aware that the same term can have different meanings in different disciplines.
- **Metacognitive skills training:** including frame identification, frame switching, and reflective pause.

Such measures could raise the level of “axiom internalization” in future generations, thereby reducing the social cost of cognitive biases.

6.7 Summary

This section has proposed a graded intervention pathway for cognitive biases caused by missing axioms.

The core idea is to move from guiding the individual to discover category confusion himself (Level 2) to explicitly internalizing the missing axiom (Level 3), and then to long-term metacognitive training for consolidation. This pathway aligns with the graded intervention protocol of Noetic Psychology and connects with educational innovation methods. For the “Sleepy” case, such intervention would likely facilitate his cognitive leap from a “probabilistic machine” to an “axiomatic machine.”

7. Testable Theoretical Hypotheses

The preceding sections have provided a dynamical interpretation of the comment-section case using the theoretical framework of Noetic Psychology and have proposed the *axiomatic pruning hypothesis*. However, this hypothesis involves neural mechanisms, temporal dynamics, and individual differences that require empirical testing. This section formalizes several conjectures raised during our discussion as explicitly testable theoretical hypotheses, serving as signposts for future research. These hypotheses are not final conclusions but a roadmap for subsequent empirical investigation.

7.1 Hypothesis 1: Distributed Storage and Retrieval of the Axiomatic System

Statement: Logical axioms (causality, non-contradiction, situation matching, rule-entity distinction, etc.) are not stored in a single brain region but are represented distributively across the prefrontal-parietal network. Different sub-axioms may be biased toward different regions (e.g., causal reasoning toward the dorsolateral prefrontal cortex, value-based judgments toward the ventromedial prefrontal cortex, concept distinction toward the parietal lobe).

Theoretical basis: Neuroimaging studies have shown that the prefrontal cortex is involved in rule learning, logical reasoning, and executive control [1,2]; the parietal lobe is involved in conceptual representation and categorical judgment [3]. This hypothesis integrates these findings into the axiomatic pruning framework.

Validation directions:

- *fMRI studies:* compare brain activation patterns during logical reasoning (e.g., syllogisms), concept distinction (e.g., “stone vs. throwing”), and causal judgment. Prediction: different axiom tasks activate different subregions of the prefrontal-parietal network, but with overlap (reflecting distributed storage).
- *Transcranial magnetic stimulation (TMS) studies:* temporarily inhibit specific prefrontal regions and observe the effect on axiom verification tasks (e.g., judging the validity of “if A then B, not B, therefore not A”). Prediction: TMS inhibition will increase error rates in axiom verification.

7.2 Hypothesis 2: Temporal Sequence of Axiomatic Pruning — Probabilistic Solution First, Axiom Verification Second

Statement: In cognitive processing, different brain regions (subconscious processors) first output probabilistic candidate solutions (intuitions, associations) based on experience; then the prefrontal system invokes axioms to prune these candidates, discarding logically invalid options. Axiom verification is a *post-hoc* process, not prior. This temporal sequence is a core prediction of the axiomatic pruning hypothesis.

Theoretical basis: The “dual-system theory” in cognitive psychology (fast System 1 intuition, slow System 2 reasoning) is compatible with this hypothesis [4]; neuroscience research on error-related

negativity (ERN) indicates that error detection signals occur after the response, suggesting post-hoc verification [5].

Validation directions:

- *Magnetoencephalography (MEG) or high-temporal-resolution EEG studies:* measure the timing of brain activity during tasks that require rule-entity distinction (e.g., quickly judging “is a stone a tool?”). Prediction: activation of sensory cortex or default-mode network precedes prefrontal activation; prefrontal “error detection” signals occur after subconscious output (approx. 300-500 ms later).
- *Behavioral experiments:* manipulate reaction time (RT), comparing error rates under fast-response vs. slow-response conditions. Prediction: under fast response (suppressing axiomatic pruning), category confusion errors increase significantly.

7.3 Hypothesis 3: Differences in Axiom Invocation Order Between Intuitive and Rational Cognitive Styles

Statement: Intuitive-inclined individuals (intuition-driven) allow subconscious probabilistic solutions to emerge first and then (attempt to) prune them with axioms, but their pruning ability is weak; rational-inclined individuals activate the axiomatic framework first, constraining probabilistic generation top-down. These differences originate from individual variations in prefrontal-sensory cortex connectivity or default-mode network function.

Theoretical basis: Behavioral genetics studies show that rational/intuitive cognitive styles are heritable and measurable [6]; neuroimaging studies have found that rational decision-makers exhibit stronger prefrontal-striatal connectivity [7].

Validation directions:

- Develop a behavioral task (e.g., a “tool categorization task”) presenting stimuli that require distinguishing physical tools from rule-based tools, asking participants to respond quickly. Simultaneously measure cognitive style using a rational/experiential inventory.
- Prediction: rational-inclined individuals will show lower error rates and, in fMRI, stronger functional connectivity between prefrontal and sensory cortices.
- *Resting-state fMRI studies:* correlate resting-state functional connectivity between prefrontal and sensory cortices with cognitive style scale scores. Prediction: rational-inclined individuals will show stronger connectivity.

7.4 Hypothesis 4: The Semantic Field Determines the Processing Order of Brain Regions

Statement: When facing a problem, the brain does not activate brain regions in a fixed sequence; instead, it prioritizes the corresponding sensory or association cortex according to the semantic field of the input (visual, linguistic, social). Axiomatic pruning then occurs uniformly after candidate solutions from all modalities have been generated.

Theoretical basis: Multimodal cognition research indicates that different sensory channels have their own dedicated primary cortices, but higher-level integration regions (e.g., prefrontal cortex) are shared across modalities [8].

Validation directions:

- *Multimodal fMRI experiments:* compare brain activation sequences between visual tasks (e.g., picture classification: which one is a tool?) and linguistic tasks (e.g., word-based concept distinction). Prediction: primary visual or language areas activate first, prefrontal later; the

activation timing does not differ across task modalities (axiomatic pruning is unified).

- *Specific experimental design* (simplified): recruit 30 healthy right-handed participants (age 20-30). Use a 2 (task modality: visual vs. auditory) $\times$ 2 (concept type: physical entity vs. abstract rule) within-subjects design.
 - Visual task: present pictures (e.g., hammer, stone, algorithm flowchart, mathematical formula). Ask: “Does the picture contain a tool?”
 - Auditory task: play spoken words (e.g., “hammer,” “stone,” “algorithm,” “rule of inference”). Ask: “Does the word refer to a tool?”
 - Record activation latencies in primary visual cortex (V1), primary auditory cortex (A1), posterior parietal cortex (concept representation), and prefrontal cortex (axiomatic pruning) using MEG.
- Expected results: primary sensory cortex activation precedes prefrontal activation (by ~100-200 ms) regardless of modality; prefrontal activation time does not differ significantly between visual and auditory tasks; no significant difference in prefrontal activation time between physical entity and abstract rule conditions.

7.5 Hypothesis 5: Positive Correlation Between Axiom Internalization and Metacognitive Capacity

Statement: Individuals who can distinguish “rule-based tools” from “physical tools” perform better on prefrontal-function integration tests (e.g., Wisconsin Card Sorting Test, cognitive flexibility tasks) and report higher metacognitive capacity.

Theoretical basis: The Wisconsin Card Sorting Test (WCST) measures cognitive flexibility and executive function, which are associated with dorsolateral prefrontal cortex function [9]. Metacognitive capacity can be measured with the Metacognitive Awareness Inventory (MAI) [10].

Validation directions:

- *Correlational study:* recruit participants of different ages and educational backgrounds. Administer:
 - Axiom internalization task (e.g., “which of the following can be created as a rule?”).
 - Wisconsin Card Sorting Test (WCST).
 - Metacognitive Awareness Inventory (MAI).
- Prediction: axiom internalization task scores will be positively correlated with WCST scores (accuracy, perseverative errors) and with MAI scores.

7.6 Hypothesis 6: Educational Intervention Can Improve Axiom Internalization

Statement: Individuals who receive systematic training in conceptual categorization and metacognitive skills (such as the graded intervention described in Section 6) will perform significantly better on axiom internalization tasks than a control group without such training.

Validation directions:

- *Quasi-experimental study:* select two similar classes or groups. One group receives a 4-week training program on “concept distinction and metacognitive skills”; the other serves as a no-treatment control. Use a pretest-posttest design measuring axiom internalization task scores.
- Prediction: the experimental group will show significantly higher posttest scores than the control group, with partial maintenance at 3-month follow-up.

7.7 Hypothesis 7: Neural Efficiency of Axiomatic Pruning Is Negatively Correlated with Mental

Health

Statement: Neural efficiency of axiomatic pruning (e.g., prefrontal activation strength and response time during pruning tasks) is negatively correlated with mental health status (e.g., anxiety, depressive symptoms). Individuals with weak axiomatic pruning function are more susceptible to subconscious bias, exhibiting more cognitive biases and psychological distress.

Validation directions:

- *Clinical study:* recruit clinically diagnosed anxiety/depression patients and healthy controls. Compare their behavioral performance and fMRI activation patterns during an axiom pruning task (e.g., category confusion judgment).
- Prediction: the patient group will show higher error rates, weaker prefrontal activation, and error rates will be positively correlated with symptom severity.

7.8 Significance of the Hypothesis List

The above hypotheses collectively form an empirical research program for the *axiomatic pruning hypothesis*. They are not independent but test different aspects (neural mechanisms, temporal dynamics, individual differences, educational intervention, clinical relevance) of the same core proposition. Confirmation of any one hypothesis will support the axiomatic pruning hypothesis; disconfirmation of any one will call for theoretical refinement.

This paper, being a theoretical contribution, does not undertake empirical testing. However, we hope these hypotheses provide actionable experimental directions for researchers in cognitive neuroscience, educational psychology, and clinical psychology, and we look forward to future empirical examination by the research community.

8. Discussion and Conclusion

8.1 Theoretical Contributions

The main theoretical contributions of this paper can be summarized as follows.

8.1.1 Proposing the Conceptual Framework of “Human Mind Alignment”

Inspired by the AI alignment problem, this paper reframes everyday cognitive biases as a *mind alignment problem* — the conflict between the probabilistic outputs of subconscious pre-trained weights and the rational demands of the current situation, leading to cognitive deviation when axiomatic pruning is absent or fails. This framework elevates cognitive bias from “individual error” to “system dynamics problem,” providing a unified perspective for understanding everyday arguments, stereotypes, logical fallacies, and related phenomena.

8.1.2 Formulating the Axiomatic Pruning Hypothesis

Based on the three-level model of consciousness in Noetic Psychology, this paper explicitly proposes the *axiomatic pruning hypothesis* and elaborates its core mechanisms: the axiomatic system is stored distributively in the prefrontal-parietal network; its function is to perform logical verification on the probabilistic candidate solutions output by the subconscious layer; pruning failure is the direct cause of cognitive bias. This hypothesis bridges macro-level psychological phenomena and micro-level neural mechanisms, offering a testable theoretical framework for cognitive psychology and neuroscience.

8.1.3 Identifying a Concrete Missing Axiom through a Case Study

Through the analysis of a comment-section conversation, this paper reveals how the absence of a specific axiom (the *rule-entity distinction axiom*) leads to cognitive bias. This case demonstrates that axiom internalization is not an abstract philosophical issue but an observable, diagnosable, and remediable real-world problem. It provides a concrete target for educational practice.

8.1.4 Providing a Graded Intervention Pathway and Testable Hypotheses

Based on the graded intervention protocol of Noetic Psychology, this paper proposes a repair pathway from Level 2 (frame optimization) to Level 3 (architecture upgrade), linking it with educational innovation methods. The seven hypotheses listed in Section 7 provide clear directions for subsequent empirical research. Thus, this paper is not only a retrospective analysis but also a forward-looking research program.

8.2 Dialogue with Existing Theories

8.2.1 Compatibility with and Extension of Dual-System Theory

Kahneman's dual-system theory (fast System 1 intuition, slow System 2 reasoning) [1] is highly compatible with our framework: subconscious probabilistic output corresponds to System 1, and axiomatic pruning corresponds to System 2.

However, this paper makes the following extensions:

- It specifies the content of System 2's "rules" — the axiomatic system (causality, non-contradiction, category distinction, etc.) — rather than a vague notion of "rationality."
- It proposes a temporal hypothesis (post-hoc verification), going beyond the simple "competition-substitution" model of dual systems.
- It links axiom internalization with neural substrates (prefrontal-parietal network), offering testable neural mechanisms.

8.2.2 Integration with Error Monitoring Theory in Cognitive Neuroscience

Neuroscience research on error-related negativity (ERN) has shown that the brain has a post-hoc error detection mechanism [2]. The axiomatic pruning hypothesis can be seen as an extension of this mechanism to the domain of logical and conceptual errors: ERN detects response errors, while axiomatic pruning detects logical errors and category confusions. The two may share common neural substrates (anterior cingulate cortex, prefrontal cortex).

8.2.3 Connection with Conceptual Change Theory in Educational Psychology

Conceptual change theory [3] studies how students move from naive theories to scientific theories. The axiom internalization hypothesis in this paper can be seen as a special case of conceptual change — shifting from a "physical-entity tool" frame to a "rule-based tool" frame. The graded intervention methods proposed here (inducing cognitive dissonance, explicit teaching, metacognitive training) are highly consistent with effective strategies identified in conceptual change research.

8.3 Limitations

8.3.1 Case-Based Limitations

This paper analyzes a single social media comment. Detailed background information about "Sleepy"

(age, education, cognitive abilities) could not be obtained, nor could follow-up intervention verification be conducted. The strength of a single-case study lies in its depth of mechanism analysis, but generalizability must be treated cautiously. Future research should employ large-sample surveys, experimental studies, or clinical studies to test the hypotheses.

8.3.2 Tentative Nature of the Theoretical Hypotheses

The axiomatic pruning hypothesis and its seven sub-hypotheses remain theoretical conjectures awaiting empirical testing by neuroimaging, behavioral experiments, and clinical studies. They should not be taken as established conclusions but as a roadmap for subsequent research.

8.3.3 Incomplete Coverage of the Axiomatic System

This paper focuses only on the *rule-entity distinction axiom*. However, the axiomatic system should include more content (causality, non-contradiction, situation matching, etc.). The internalization mechanisms, neural substrates, and developmental trajectories of different axioms may differ, requiring further investigation.

8.3.4 Consideration of Cultural Differences

The case analyzed comes from Chinese social media, and the analysis relies on the semantic category of the Chinese word “tool” (*gongju*). Different languages and cultures may conceptualize “tool” differently. Is axiom internalization culturally influenced? Is the rule-entity distinction universal across cultures? These questions await cross-cultural research.

8.4 Practical Implications

8.4.1 Implications for Educational Practice

This analysis shows that category confusion is an important source of cognitive bias, yet current education pays insufficient attention to category distinction training. Recommendations:

- Introduce basic logic and concept analysis courses in primary and secondary education.
- Strengthen “concept definition” and “frame identification” training in higher education, especially in the humanities and social sciences.
- Develop metacognitive skills training tools (e.g., reflective pause, frame-switching exercises).

8.4.2 Implications for Public Discourse

Arguments on social media often become deadlocked due to category confusion. This analysis suggests that when a debate cannot move forward, participants can:

- Pause the factual debate and first clarify the definition of the core concept.
- Ask the other party: “Do you mean ‘tool’ as a physical entity or as a rule/method?”
- Reflect on whether they themselves are confusing different categories.

8.4.3 Implications for Psychotherapy

For cognitive biases rooted in concept confusion (e.g., some forms of anxiety, obsessive thinking), therapists can:

- Identify the specific category confusion (e.g., “is failure an event or an identity?”).
- Guide the client to distinguish between entity and rule, fact and value, description and norm.
- Use Socratic questioning and explicit teaching to help internalize the missing distinction axiom.

8.5 Future Research Directions

Based on the theoretical framework and hypotheses of this paper, future research can proceed in the following directions:

- **Neural mechanism validation:** use fMRI, MEG, TMS, etc., to test the hypotheses about distributed storage, temporal sequence, and individual differences in axiomatic pruning.
- **Educational intervention studies:** design and implement concept-categorization training courses, using randomized controlled trials to evaluate their effects on axiom internalization and reduction of cognitive biases.
- **Cross-cultural comparisons:** compare performance on rule-entity distinction tasks across different languages and cultures to test the universality of the axioms.
- **Clinical translation:** develop axiom-pruning-based training programs to assist in treating psychological disorders related to concept confusion (e.g., some forms of obsessive-compulsive disorder, borderline personality disorder).
- **Bidirectional inspiration with AI alignment research:** apply findings from human axiom internalization to AI axiom injection design; conversely, use AI engineering implementations (e.g., the “Parliament of Logical Subuniverses”) as computational models for understanding the human mind.

8.6 Conclusion

Starting from a brief comment-section conversation and applying the theoretical framework of Noetic Psychology, this paper has revealed a common but rarely theorized cognitive phenomenon: due to the failure to internalize the *rule-entity distinction axiom*, the probabilistic outputs of subconscious pre-trained weights bypass axiomatic pruning and directly dominate cognitive judgment, producing category confusion.

We have proposed the concept of *human mind alignment*, structurally analogized it to the AI alignment problem, and argued that both share the same dynamical logic. We have further formulated the *axiomatic pruning hypothesis*, specifying the neural distribution of the axiomatic system, the temporal sequence of pruning, and the mechanisms of pruning failure, and have listed seven testable sub-hypotheses. Finally, based on the graded intervention protocol of Noetic Psychology, we have outlined a repair pathway for axiom internalization, providing a theoretical foundation for educational practice and psychotherapy.

This analysis shows that many impasses in everyday arguments stem not from factual disagreement or lack of logical ability, but from missing internalized axioms and failed pruning processes. Repairing these biases does not require “implanting correct answers” but rather helping individuals internalize the missing axioms and restore their own capacity for rational judgment. This is precisely the practical embodiment of the “free will centrism” principle in Noetic Psychology — the therapist does not make judgments for the individual but helps the individual acquire the ability to make correct judgments.

When “Sleepy” can finally spontaneously say, “I confused the physical material with the operational rule,” he will have accomplished not an admission of error but a cognitive leap of axiom internalization. This is what this paper envisions: from a “probabilistic machine” to an “axiomatic machine” — a transition that applies not only to AI but also to every human mind seeking rational cognition.

This paper is an application demonstration of the Noetic Psychology theoretical framework to the analysis of micro-level cognitive biases. Further applications await future research.

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